

1. Executive Summary

This report assesses the accuracy of **main.py (v1)**, a Random Forest regression model trained on a multi-domain Welsh Government dataset containing diagnostic waiting times, emergency department attendances, cancer pathways, and RTT waiting list metrics. The purpose of the analysis is to evaluate how well the model predicts the numeric *value* field using categorical descriptors and engineered date features.

The model was trained using one-hot encoded predictors derived from **description**, **geography**, **sex**, **age**, **source_file**, and engineered **year** and **month** variables. The dataset includes a wide range of health-system indicators, such as:

“diagnostics pathway waiting up to 8 weeks” (e.g., 63,319 on 2019-01-01) “diagnostics pathway waiting over 8 weeks” (e.g., 3,992 on 2019-01-01) “diagnostics pathway total waiting” (e.g., 67,311 on 2019-01-01)

The model was evaluated using an 80/20 train-test split. The code computes the **Mean Absolute Error (MAE)**; however, the numerical MAE result is not provided in the inputs.

Key findings

- The model successfully trains and produces predictions without errors.
- The dataset contains strong temporal structure (monthly data from 2019–2025), which the model can partially exploit through engineered year/month features.
- The dataset is **trimmed**, meaning the model is trained on incomplete historical context, which likely reduces predictive accuracy.
- The model uses only categorical descriptors and date parts; no domain-specific numerical features are included, limiting explanatory power.

Overall conclusion The model is functional and produces an MAE score, but its accuracy cannot be fully assessed without the missing MAE value. The trimming of the dataset and the simplistic feature engineering approach constrain the model’s predictive performance.

2. Background and Purpose

The Welsh Government modelling team requested an assessment of the accuracy of **main.py (v1)**. The model aims to predict the numeric *value* associated with various health-system metrics (e.g., waiting times, attendances, treatment counts) using categorical descriptors and date-based features.

The purpose of this report is to:

- Summarise the modelling approach used in **main.py (v1)**.
- Evaluate the model’s predictive performance.

- Identify limitations arising from the dataset and modelling choices.
- Provide an evidence-based interpretation suitable for both technical and executive audiences.

3. Methodology Summary

The methodology is fully defined in the provided code:

Data preparation

- The dataset is loaded from *training_dataset.csv*.
- The target variable **value** is converted to numeric.
- The **date** field is parsed and split into **year** and **month**.
- Predictors include:

“description”, “geography”, “sex”, “age”, “source_file”, “year”, “month”

- All predictors are one-hot encoded using `pd.get_dummies`.

Row filtering

Rows are retained only where both **value** and **date** are valid.

Train/test split

- 80% training
- 20% testing
- `random_state=42` ensures reproducibility.

Model

- RandomForestRegressor
- 200 trees
- Default hyperparameters otherwise.

Evaluation

- Predictions are generated on the test set.
- **Mean Absolute Error (MAE)** is computed via `mean_absolute_error(y_test, preds)`.

Missing information

- **[INFORMATION NEEDED: The actual MAE value printed by the script.]**

4. Key Findings

4.1 Model behaviour

- The model trains successfully using the trimmed dataset.
- The Random Forest approach is robust to categorical one-hot encoded features and can model non-linear relationships.

4.2 Dataset characteristics

The dataset contains monthly values for a wide range of health indicators. Examples include:

- *“diagnostics pathway waiting up to 8 weeks”*

“63319” on “2019-01-01”

- *“diagnostics pathway waiting over 8 weeks”*

“3992” on “2019-01-01”

- *“diagnostics pathway total waiting”*

“67311” on “2019-01-01”

These values are directly quoted from the dataset.

4.3 Predictive accuracy

- The model outputs an MAE score, but the numerical value is not included in the inputs.
- Without the MAE, accuracy cannot be quantified.

4.4 Feature limitations

- The model does not use lagged features, rolling averages, or domain-specific engineered variables.
- All predictors except year/month are categorical, limiting the model’s ability to capture temporal dynamics.

4.5 Impact of trimmed data

- The dataset is explicitly described as **trimmed**, meaning historical context is incomplete.
- This reduces the model’s ability to learn long-term patterns.

5. Limitations and Caveats

This section intentionally does not water down limitations.

Data limitations

- **Trimmed dataset:** The dataset is incomplete, reducing the model's ability to learn long-term seasonal or structural patterns.
- **Heterogeneous indicators:** The dataset mixes fundamentally different metrics (e.g., ED attendances, cancer treatment counts, RTT waiting lists). A single model predicting all of them may underperform.
- **No validation of data quality:** The code does not check for outliers, structural breaks (e.g., COVID-19), or missingness patterns.

Methodological limitations

- **No hyperparameter tuning:** The Random Forest uses default settings except for number of trees.
- **No temporal modelling:** The model does not incorporate autoregressive or sequential dependencies.
- **No feature selection:** All one-hot encoded variables are included, potentially introducing noise.
- **No cross-validation:** Only a single train/test split is used.

Evaluation limitations

- **MAE value missing:**

[INFORMATION NEEDED: The actual MAE printed by the script.]

- Without the MAE, accuracy cannot be interpreted.

6. Recommendations

Recommendations are in scope because the purpose is to assess model accuracy.

Short-term recommendations

- **Provide the MAE value** so accuracy can be fully assessed.
- **Segment the dataset** by indicator type (e.g., diagnostics, ED, RTT) and train separate models.
- **Add temporal features**, such as lagged values or rolling means.

Medium-term recommendations

- **Introduce cross-validation** to stabilise accuracy estimates.
- **Perform hyperparameter tuning** (e.g., RandomizedSearchCV).
- **Assess feature importance** to understand model behaviour.

Long-term recommendations

- Consider **time-series models** (e.g., Prophet, ARIMA, LSTM) for indicators with strong temporal structure.

7. Appendix

Quoted lines from the dataset (as required)

Examples of directly quoted dataset entries:

“63319,” “diagnostics pathway waiting up to 8 weeks”, “2019-01-01” “3992,”
“diagnostics pathway waiting over 8 weeks”, “2019-01-01” “67311,” “diagnostics
pathway total waiting”, “2019-01-01”